Improved Small Object Detection with YOLOv8 in Remote Sensing Imagery: Architecture Modifications and Loss Function Optimization

1st Nurul Amelia
School of Electrical Engineering
Center of Excellence AILO
Telkom University
Bandung, Indonesia
nramelia@student.telkomuniversity.ac.id

2nd Suryo Adhi Wibowo
School of Electrical Engineering
Center of Excellence AILO
Telkom University
Bandung, Indonesia
suryoadhiwibowo@telkomuniversity.ac.id
(Corresponding author)

3rd Casi Setianingsih
School of Electrical Engineering
Center of Excellence AILO
Telkom University
Bandung, Indonesia
setiacasie@telkomuniversity.ac.id

2024 7th International Conference on Information and Communications Technology (ICOIACT) | 979-8-3315-3620-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICOIACT64819.2024.10913393

***Abstract*—Remote sensing's ability to identify small objects is essential for earth observation, but algorithms developed for remote sensing remain challenged in achieving high accuracy. This research addresses critical challenges in remote sensing, including the small size of objects relative to the entire image, random object orientations requiring rotation-invariant detection, and significant variations in object scale. Furthermore, the model effectively handles noise occlusions and demonstrates the effectiveness of the Wise Intersection over Union (WIoU) loss function in improving performance and robustness. This research proposed a modified YOLOv8 model by replacing the C2f layer with the C3 layer, adding layers to the neck, and adding a head component. However, the modified model underperforms when using the Complete Intersection over Union (CIoU) loss function, achieving mean Average Precision (mAP) of 76.146% accuracy due to increased complexity, causing noisier predictions. In contrast, applying the WIoU loss function improves the model's performance, achieving an mAP of 77.183% accuracy and an improvement of 0.875% over the base model by better handling prediction variability.**

***Index Terms*—small object detection, YOLOv8, C3, head, WIoU**

I. Introduction

Remote sensing images are utilized in a variety of fields, such as traffic monitoring [1], town planning [2], intelligent monitoring [3], and environmental monitoring [4]. Advancements in deep learning technology have analyzed remote sensing images more efficiently and intelligently, minimizing the reliance on manual processes. As essential tasks in computer vision, object recognition and detection are crucial for extracting meaningful insights from these images [1].

In remote sensing, images are captured using various technologies such as satellites, aircraft, and, more recently, drone-based sensors. As a result, gathering training data for remote sensing is more challenging. However, the unique remote sensing image capture methods introduce several vital challenges. Firstly, objects within these images are often relatively small compared to the overall image size, which limits the amount of feature information available for detection. For instance, an object might only cover a few dozen pixels in an image with millions of pixels. Second, because objects might appear at different angles, object detection algorithms must provide characteristics invariant to rotation to improve detection accuracy [5]. The size of objects can vary significantly both within a one image and divergent images, such as the difference in scale between a car and an airplane [6], [7], [8], [9], [10], [11]. Object detection algorithms must also address noise and occlusions [12].

Deep learning techniques have proven to be highly effective in this domain and have grown to be an important subject of study [13], [14]. These techniques generally fall into two categories: supervised and unsupervised. Unsupervised learning involves extracting patterns or high-level semantics from unlabeled data. In contrast, Convolutional Neural Networks (CNNs) have become a preferred choice for supervised object detection across various image processing domains. Popular YOLO models have demonstrated outstanding performance [15], [16], [17], as they demonstrate a strong capability to simultaneously learn regions of interest and contextual information. YOLOv8 [18] has demonstrated efficiency and accuracy, achieving high mean average precision. Even though it does not explicitly target small object detection, remains a robust benchmark method [19], [20].

This research presents a proposed YOLOv8-based object detection architecture for remote sensing images trained on the DIOR dataset. This research outlines the key contributions as follows:

1) Firstly, replacing C2f with C3 in P4 and P6 from YOLOv5 can enhance feature fusion and improve the feature map's representation capability.
2) Secondly, seven stages were added to the neck, specifically stages P16–P18 and P25–P27. Additionally, one detection layer was added to the head, increasing the number of detection heads to four. This enhancement improves the system's ability to handle targets at various scales, including tiny, small, medium, and large.

3) Finally, optimize the loss functions using WIoU [21] [22].

The proposed architecture in this research is shown in Fig. 1.

II. Related Work

Remote sensing images frequently exhibit significant variances in target scale and are influenced by complicated environmental elements such as lighting and weather. These challenges make it difficult to achieve optimal performance when applying standard deep learning models directly to these images [23]. In response to these challenges, researchers often adjust R-CNN and YOLO-based models to meet the specific requirements of remote sensing applications [24].

Xu et al. [25] developed an improved remote sensing object detection framework on the basic of YOLOv3. This framework incorporates a feature enhancement network designed to extract essential features more effectively. When tested on remote sensing datasets, the model successfully detected targets.

Liu et al. [26] proposed an improved model on the basic of YOLOv8, called YOLO-SSP, which enhances detection accuracy by optimizing the downsampling layers to capture finer details and utilizing hierarchical pooling operations to generate weights from different spatial locations. Despite various efforts to enhance multi-scale object detection, accurately identifying tightly clustered small targets in remote sensing imagery remains a significant challenge.

Liu et al. [27] introduced the YOLO-extract method, inspired by YOLOv5, which utilizes residual concepts to enhance feature extraction capabilities. This model integrates the coordinate attention mechanism and mixed dilated convolution. To speed up model convergence, Focal-α EIoU was used to replace the CIoU loss function. Additionally, an extra detection head was incorporated to specifically target small and densely packed objects.

Zhang et al. [28] enhanced the backbone network's receptive field and feature fusion components based on YOLOv5, improving detection performance by strengthening the feature representation of small objects. However, enhanced multiscale feature extraction typically results in more complex model architectures and increased computational demands, which can reduce the model's inference speed. Some approaches aim to increase the model's consideration to small objects by refining the distance metrics or adjusting the bounding box thresholds.

III. Material And Research Method

A. Dataset

The DIOR dataset [3] used in this study consists of 20 object categories. The earth observation community considers it one of the most extensive, diverse and publicly available databases for object detection. The dataset consists of 23,463 high-resolution remote sensing images, in addition to 192,472 object instances, all of which were manually annotated with axis-aligned bounding boxes. The dataset features 20 commonly observed object categories and shows substantial variation in object size, reflecting differences in sensor spatial resolution and size differences between and within categories. In addition, DIOR offers a wide diversity of images, which is crucial for developing robust object detection models. In comparison, many other datasets require more variety in their images, partially or entirely. In addition, DIOR provides significant challenges for detection algorithms due to the high inter-class similarity and considerable variation within each class.

B. C3

The C2 module in YOLOv8 [18] refers to the Cross Stage Partial (CSP) Bottleneck with two convolutions, which serves as a key building block in the architecture. The C2f module is a more efficient version of the C2 module, designed to enhance the model's execution speed while maintaining comparable performance. This improvement is achieved through specific adjustments made to the original C2 module. The C2f block enhances execution speed by dividing the feature map produced by the initial convolutional block. One part of the feature map is routed through the Bottleneck block, while the other bypasses the Bottleneck and goes directly to the Concat block.

The cross-stage partial block consisting of three convolutional layers with jump connections is referred to as the C3 layer. The C3 layer divides in two the input feature map: one part will be directed straight to the output, while the other part will go through multiple convolutional layers with e times the number of output channels (ch_out). The output feature map is then generated by merging the two parts along the channel dimensions. The C3 layer leverages cross-stage partial connections to reduce network complexity and redundancy. The merged edges in the C3 block represent features extracted from multiple feature map layers, resulting in a more comprehensive representation that captures both high-level and low-level information.

C. Wise Intersection Over Union

The bounding box regression loss function is an essential component in object detection, especially in small detection objects. Yu et al. [29] were the first to introduce the Intersection over Union (IoU) loss function to predict bounding boxes. During training, the model generates a prediction box and a ground truth box, and the IoU metric measures the difference between them, measuring the model's accuracy.

The bounding box regression loss function is crucial in measuring the positional difference between the predicted and ground truth boxes. This process helps optimize the model by adjusting the predicted box's position, ultimately enhancing the model's accuracy. Researchers have developed and tested numerous loss functions, including Distance-IoU (DIoU) and Complete-IoU (CIoU) proposed by Zheng et al. [30], as well as Generalized-IoU (GIoU) introduced by Rezatofighi et al. [31]. CIoU Loss assesses the alignment between bounding boxes by measuring the distance between their centres, the degree of overlap, and their size. It also includes an aspect ratio term to address any discrepancies between the predicted and actual aspect ratios [21].

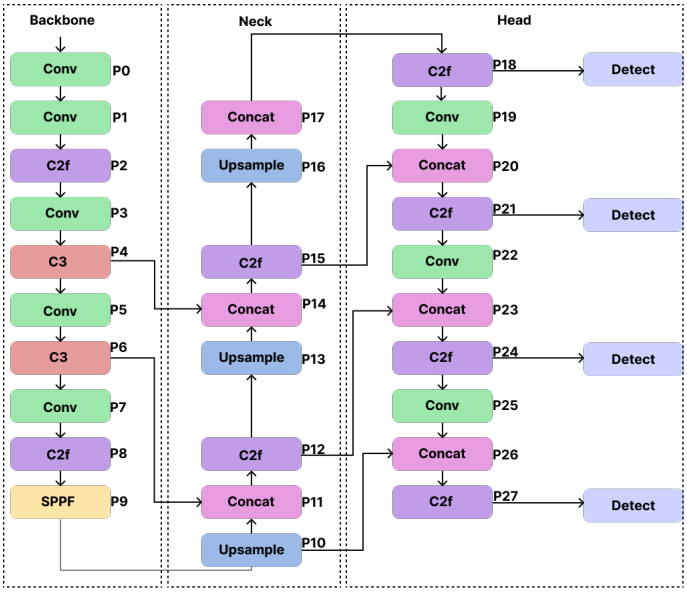


Fig. 1. The structure modification of YOLOv8.

CIoU Loss uses a monotonic focus technique which assumes high-quality training data to achieve improved bounding box prediction accuracy. However, in real-world scenarios, datasets often contain variations and inaccuracies in annotations. An optimized loss function paired with low-quality annotations can disrupt feature learning, delay convergence, and weaken the model's robustness. Wise-IoU is used as the regression loss function to overcome these concerns, improving the model's generalisation capacity and speeding up the convergence process [21].

Wise-IoU presents a dynamic non-monotonic focus mechanism that adopts gradient gains based on how much each bounding box prediction deviates from the ground truth. The non-monotonic behavior means that the gradient gains do not change in a straightforward way as the loss increases. This allows the model to reduce the influence of low-quality predictions while also slightly lowering the gradient gains for highly accurate predictions, ensuring a more balanced learning process and better overall performance [21].

The IoU loss can be described as Equation (1):

$$L_{\text{IoU}} = 1 - \text{IoU} = 1 - \frac{w_i h_i}{wh + w_g h_g - w_i h_i} \tag{1}$$

$$\frac{\partial \mathcal{L}_{\text{IoU}}}{\partial w_i} = \begin{cases} -\frac{h_i(1+\text{IoU})}{wh+w_{gt}h_{gt}-w_i h_i}, & w_i > 0 \\ 0, & w_i = 0 \end{cases} \tag{2}$$

$$L_{\text{WIoUv1}} = R_{\text{WIoU}} L_{\text{IoU}} \tag{3}$$

According to the developer of WIoU [22], this loss function integrates a dynamic non-monotonic focusing mechanism. Unlike traditional IoU-based algorithms, WIoU can dynamically adjust the loss coefficient for each training sample. The loss coefficient is reduced for more straightforward samples, allowing the model to shift its focus away from these samples. Conversely, the loss coefficient is increased for more challenging samples, enabling the model to pay closer attention to complex examples. This approach helps the model learn from more challenging samples more effectively, ultimately enhancing overall detection performance.

WIoU also applies an outlier factor, denoted as β, to consider of anchor boxes. When β is higher, the model provides a lower gradient gain to keep low-quality input from interrupting the training process. This adaptive adjustment stabilizes the training and enables the model to learn effectively from both high- and low-quality data.

Within this mechanism, the non-monotonic focus coefficient as expresed in Equation (4):

$$r = \frac{\beta}{\delta \alpha^{\beta} - \delta} \tag{4}$$

Where the parameter $\beta = \frac{\mathcal{L}^{*}_{\text{IoU}}}{\overline{\mathcal{L}}_{\text{IoU}}} \in [0, +\infty]$ quantifies the outlier degree of the predicted bounding box quality. To align with the outlier calculation strategy, WIoU introduces the non-monotonic focusing coefficient r derived from β. This adjustment forms the foundation of WIoUv3, as expressed in Equation (5).

$$L_{\text{WIoUv3}} = r \cdot L_{\text{WIoUv1}} \tag{5}$$

IV. Results And Discussion

In this research, in the data preparation and preprocessing stage, the system used was a computer running Windows 11, equipped with an Intel Core i5 processor. This stage involved preparing the data and converting files from .xml to .text format, which would be used for model training. Once the data preparation was completed, the model training

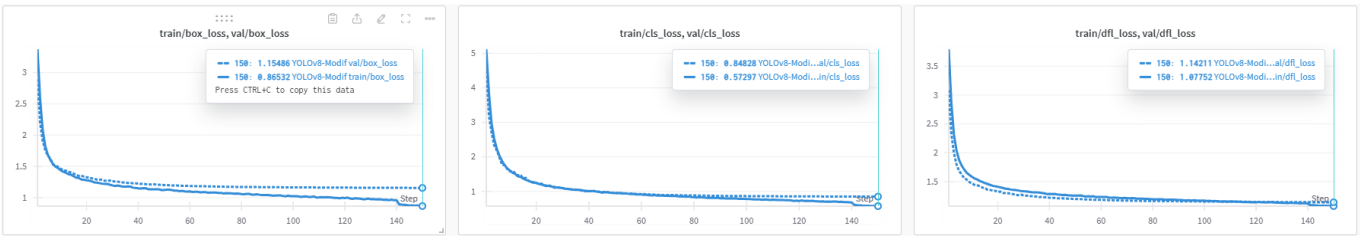


Fig. 2. The results modification of YOLOv8.



Fig. 3. Small object detection results using YOLOv8-Modif.

process was carried out using an NVIDIA A100-SXM4-40GB graphics processing unit (GPU), which offers high capacity and performance to handle intensive computations required for training artificial intelligence models. This hardware allows for faster and more efficient model training, especially when managing large and complex datasets.

Before modifying the model architecture, the researcher conducted experiments to set suitable hyperparameters for this research. The adjusted hyperparameters include batch size, epoch, image size, optimizer, and learning rate. The hyperparameter settings ensure that the model is trained with optimal parameters to achieve the best results. After conducting several experiments to set the hyperparameters and analyze the results, the researcher identified the hyperparameter combination that resulted in the best performance. Further modifications were made to the YOLOv8 architecture to improve the accuracy and efficiency of the model. The hyperparameters used in the training process are as follows: batch size = 8, epoch = 150, image size = 840, optimizer = 'SGD', and learning rate = 0.01. These settings were chosen based on the evaluation of the previous experiments, where this combination provided the most optimal training results for the context of this research. Table I shows the training results of YOLOv8-Base and YOLOv8-Modif after setting the hyperparameters, modifying the architecture and optimizing the loss function.

Table 1 compares the performance of the modified YOLOv8 model to that of the basic model. The differences may be explained by architectural changes and the characteristics of the loss functions, CIoU and WIoU. The architecture becomes more complex when replacing the C2f layer with the C3 layer, adding more layers to the neck, and adding a head part that becomes four. However, this complexity negatively affects performance if the loss function is not well-suited to the modified structure.

TABLE I
TRAINING RESULTS FOR MODEL COMPARISON

Model	CIoU	WIoU
YOLOv8-Base	76.309%	76.188%
YOLOv8-Modif	76.146%	**77.183%**

The CIoU loss function focuses on the proper geometric alignment between the predicted bounding box and the ground truth. This function will consider the distance between centres, overlapping areas, size, and aspect ratio of bounding boxes. While CIoU can be effective for high-quality datasets with proper annotations, it assumes that the architectural components provide highly accurate predictions. In the modified YOLOv8 architecture, the deeper C3 layers, additional neck layers, and the addition of head layers introduce more parameters and complexity. This leads to noisier predictions and difficulty achieving tight bounding box alignment. As a result, the CIoU loss function may penalize minor deviations, leading to performance degradation compared to the base model.

On the other hand, the WIoU loss function provides a more adaptive mechanism for handling both high-quality and low-quality predictions. Unlike CIoU, which follows a monotonic focus mechanism, WIoU incorporates a dynamic non-monotonic focus mechanism. This mechanism adjusts the gradient gains on the basic of the quality of the predicted bounding boxes, dynamically balancing the contributions of low-quality and high-quality predictions. With the modified

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON DIOR.

	CSFF [32]	PANet [33]	YOLOv8-Base(C)	YOLOv8-Base(W)	YOLOv8-Modif(C)	YOLOv8-Modif(W)
Backbone	ResNet-101	ResNet-50	CSPDarknet	CSPDarknet	CSPDarknet	CSPDarknet
Airplane	0.572	**0.624**	0.620	0.610	0.600	0.620
Airport	0.796	0.763	**0.90**	0.880	0.870	0.890
Baseball field	0.701	0.716	0.700	0.690	**0.720**	0.700
Basketball court	0.874	0.873	0.890	0.890	0.880	**0.910**
Bridge	0.461	**0.486**	0.390	0.390	0.390	0.430
Chimney	0.766	**0.793**	0.760	0.760	0.760	0.770
Dam	0.627	0.655	0.670	**0.730**	0.710	0.720
Expressway service area	0.826	0.759	0.830	**0.870**	0.850	0.860
Expressway toll station	**0.732**	0.728	0.610	0.580	0.630	0.630
Golf field	0.782	0.764	0.840	0.840	0.840	**0.880**
Ground track field	0.816	0.825	0.820	0.810	**0.840**	0.820
Harbor	0.507	0.472	0.720	**0.730**	0.700	0.710
Overpass	0.595	0.606	**0.610**	0.580	0.600	0.600
Ship	0.733	0.720	0.890	0.880	0.900	**0.900**
Stadium	0.634	0.687	0.710	0.710	0.730	**0.750**
Storage tank	0.585	0.626	0.710	0.690	0.730	**0.770**
Tennis court	0.859	0.812	0.880	0.880	**0.880**	0.870
Train station	0.619	0.563	0.760	**0.770**	0.670	0.680
Vehicle	0.429	**0.505**	0.330	0.340	0.380	0.400
Windmill	0.869	0.880	0.880	0.870	0.870	**0.880**
mean Average Precision (mAP) (%)	68.000%	69.300%	76.309%	76.188%	76.146%	**77.183%**

YOLOv8 architecture, which introduces additional complexity, WIoU's adaptive nature becomes advantageous. It decrease the influence of low-quality predictions while allowing the model to learn effectively from high-quality ones. This helps the modified architecture converge more efficiently and achieve better generalization.

Table I compares the training performance between the YOLOv8-Base and YOLOv8-Modif models using the CIoU and WIoU metrics. The YOLOv8-Base model achieves 76.309% for CIoU and 76.188% for WIoU. The YOLOv8-Modif model slightly lowers CIoU to 76.146% but improves WIoU performance to 77.183%. The modified version indicates enhanced weighted IoU performance, reflecting better bounding box regression and improved detection accuracy.

Table II details and compares the experimental results across different object classes with previous research. First, the experimental values of CIoU and WIoU for the YOLOv8-Base and YOLOv8-Modif models, in most classes, the YOLOv8-Modif model using WIoU achieves higher results compared to using CIoU, which indicates the performance of WIoU for small object detection is better for particular objects. The YOLOv8-Modif model specifically improves performance by detecting Bridges, Dams, Toll Road service areas, Stadiums, Storage tanks, and Vehicles. However, both models have similar performance or slight variations in CIoU and WIoU for some object classes, such as Chimneys, Tennis Courts, and Windmills. Overall, the YOLOv8-Modified model shows improved performance in WIoU while maintaining competitive results in CIoU, indicating that the modification positively contributes to a more balanced and robust detection of small objects. In addition, compared to previous studies, some classes perform better in previous studies compared to YOLOv8-Modif, as shown in Table II.

Fig. 2 shows the training and validation losses for box loss, classification loss (cls loss), and distribution focal loss (DFL loss), which offer insights into the learning process and model convergence. In the case of box loss, the training and validation curves demonstrate a rapid decrease in the initial training stage. This indicates that the model is quickly learning to predict the bounding box more accurately. Around 20 steps into the process, the curves begin to flatten, indicating that the model has reached a point of convergence. By the 150th step, the training loss stabilized at around 0.86532. In contrast, the validation loss settles slightly higher at 1.15486, with a slight difference between the two, thus indicating a well-generalized model that does not show significant signs of overfitting.

Similarly, the classification loss follows a comparable pattern. The steep decline in loss early on reflects the model's rapid improvement in object classification. By the 150th step, the training classification loss reaches 0.57297, and the validation loss follows closely at 0.84828. Although there is a gap between the training and validation losses, the gap is relatively small and indicates that the model is generalizing reasonably well. However, further tuning might be necessary to minimize the difference.

The DFL loss exhibits a slightly slower convergence than the box and classification losses. The curves continue to decrease steadily throughout the training process, with the final training loss reaching 1.07752 and the validation loss stabilizing at 1.14211. While the DFL loss shows slower convergence, the overall behaviour is consistent with the expected learning trend, and the gap between the training and validation losses remains relatively small, signalling effective learning without overfitting.

Fig. 3 is small object detection results on satellite imagery highlight various structures, such as chimneys with confidence scores of 0.93 and 0.94, windmills with scores of 0.70 and 0.65, a stadium with a score of 0.77, and overlapping fields

identified as a ground track field 0.75 and a baseball field 0.77.

V. CONCLUSION

The reseacher conducted several experiments in this research and achieved the best performance with 77.183% accuracy. This research was improved by modifying the YOLOv8n architecture, especially in the backbone, neck, and detection components, and optimizing the loss function using WIoU. The modified YOLOv8 model showed an improvement of 0.875% over the base YOLOv8 model, which achieved an accuracy of 76.309%. Future research is encouraged to increase the detecting performance, especially for the Bridge and Vehicle classes, as the confusion matrix value is still below 0.50, and to explore experiments with the latest YOLO models, such as YOLOv10 and YOLOv11.

ACKNOWLEDGMENT

The authors would like to express our sincere gratitude to the Center of Excellence Artificial Intelligence for Learning and Optimization (CoE AILO), support by the RIIM Batch 3 program grant funded by LPDP, the Ministry of Finance, Rep. Dr. Ir. Indonesia, and BRIN (no. 81/IV/KS/05/2023) and supported by Telkom University. Their guidance and resources made this work possible.

REFERENCES

[1] S. Xie, M. Zhou, C. Wang and S. Huang, "CSPPartial-YOLO: A Lightweight YOLO-Based Method for Typical Objects Detection in Remote Sensing Images," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, pp. 388-399, 2024, doi: 10.1109/JSTARS.2023.3329235.

[2] X. Lu, J. Ji, Z. Xing and Q. Miao, "Attention and Feature Fusion SSD for Remote Sensing Object Detection," in IEEE Transactions on Instrumentation and Measurement, vol. 70, pp. 1-9, 2021, Art no. 5501309, doi: 10.1109/TIM.2021.3052575.

[3] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," ISPRS Journal of Photogrammetry and Remote Sensing, vol. 159, pp. 296-307, 2020, doi: 10.1016/j.isprsjprs.2019.11.023.

[4] L. Shen, B. Lang and Z. Song, "CA-YOLO: Model Optimization for Remote Sensing Image Object Detection," in IEEE Access, vol. 11, pp. 64769-64781, 2023, doi: 10.1109/ACCESS.2023.3290480.

[5] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," IEEE Transactions on Geoscience and Remote Sensing, vol. 56, no. 4, pp. 2337–2348, 2017.

[6] T.-Y. Lin, P. Doll´ ar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in Proc. IEEE Conf. Comput. Vision Pattern Recogn., Honolulu, HI, Jun. 2017, pp. 2117–2125.

[7] M. Ju, H. Luo, Z. Wang, B. Hui, and Z. Chang, "The application of improved YOLO V3 in multi-scale target detection," J. Appl. Sci., vol. 9, no. 18, pp. 3774–3375, Jan. 2019.

[8] Y. Ren, C. Zhu, and S. Xiao, "Small object detection in optical remote sensing images via modified Faster R-CNN," J. Appl. Sci., vol. 8, no. 5, pp. 813:1–813:11, 2018.

[9] M. Mandal, M. Shah, P. Meena, S. Devi, and S. K. Vipparthi, "Avdnet: A small-sized vehicle detection network for aerial visual data," IEEE Geosci. Remote Sens. Lett., pp. 494–498, 2019.

[10] M. Mandal, M. Shah, P. Meena, and S. K. Vipparthi, "SSSDET: Simple Short and Shallow Network for Resource Efficient Vehicle Detection in Aerial Scenes," in Proc. IEEE Int. Conf. Image Process., Taipei, Taiwan, Sep. 2019, pp. 3098–3102.

[11] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multiscale object detection in remote sensing imagery with convolutional neural networks," ISPRS journal of photogrammetry and remote sensing, vol. 145, pp. 3–22, 2018.

[12] M. Sharma et al., "YOLOrs: Object Detection in Multimodal Remote Sensing Imagery," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 14, pp. 1497-1508, 2021, doi: 10.1109/JSTARS.2020.3041316.

[13] Y. Sun, Z. Wang, X. Sun, and K. Fu, "Span: Strong scattering point aware network for ship detection and classification in large-scale SAR imagery," IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens., vol. 15, pp. 6083–6101, 2022.

[14] Z. Honget al., "Multi-scale ship detection from SAR and optical imagery via a more accurate yolov3," IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens., vol. 14, pp. 6083–6101, 2021.

[15] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit., 2016, pp. 779–788.

[16] J. Redmon and A. Farhadi, "Yolo9000: Better, faster, stronger," in Proc. IEEEComput.Soc.Conf.Comput.Vis.PatternRecognit., 2017, pp. 7263-7271.

[17] A. Farhadi and J. Redmon, "Yolov3: An incremental improvement," in Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit., 2018, pp. 1804–1804.

[18] G. Jocher, et al., "Ultralytics/YOLOv8," GitHub repository, https://github.com/ultralytics/yolov8, 2023.

[19] A. Bochkovskiy, C. -Y. Wang, and H. -Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," 2020, arXiv:2004.10934.

[20] Q. Xu, Y. Li and Z. Shi, "LMO-YOLO: A Ship Detection Model for Low-Resolution Optical Satellite Imagery," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 15, pp. 4117-4131, 2022, doi: 10.1109/JSTARS.2022.3176141.

[21] L. Shen, B. Lang and Z. Song, "DS-YOLOv8-Based Object Detection Method for Remote Sensing Images," in IEEE Access, vol. 11, pp. 125122-125137, 2023, doi: 10.1109/ACCESS.2023.3330844.

[22] Z. Tong, Y. Chen, Z. Xu, and R. Yu, "Wise-IoU: Bounding box regression loss with dynamic focusing mechanism," 2023, arXiv:2301.10051.

[23] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning based change detection in remote sensing images:Areview,"Remote Sens., vol. 14, no. 4, p. 871, 2022.

[24] H. Yi, B. Liu, B. Zhao and E. Liu, "Small Object Detection Algorithm Based on Improved YOLOv8 for Remote Sensing," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, pp. 1734-1747, 2024, doi: 10.1109/JSTARS.2023.3339235.

[25] D. Xu and Y. Wu,"Fe-YOLO:A feature enhancement network for remote sensing target detection," Remote Sens., vol. 13, no. 7, 2021, Art. no. 1311.

[26] Y. Liu, D. Yang, T. Song, Y. Ye, and X. Zhang, "YOLO-SSP: An object detection model based on pyramid spatial attention and improved downsampling strategy for remote sensing images," Vis. Comput., vol. 40, pp. 1–18, 2024.

[27] Z. Liu, Y. Gao, Q. Du, M. Chen, and W. Lv, "YOLO-extract: Improved YOLOv5 for aircraft object detection in remote sensing images," IEEE Access, vol. 11, pp. 1742–1751, 2023, doi: 10.1109/ACCESS.2023.3233964.

[28] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCAY-OLO for small object detection in remote sensing images," IEEE Trans. Geosci. Remote Sens., vol. 62, 2024, Art. no. 5611215, doi: 10.1109/TGRS.2024.3363057.

[29] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox," in Proc. 24th ACM Int. Conf. Multimedia, Oct. 2016, pp. 516–520, doi: 10.1145/2964284.2967274.

[30] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," 2019, arXiv:1911.08287.

[31] H. Rezatofighi, N. Tsoi, J. Y. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2019, pp. 658–666.

[32] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," Commun. ACM, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.

[33] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," IEEE Geosci. Remote Sens. Lett., to be published, doi: 10.1109/LGRS.2020.2975541.